

9 March 2001

Teaching Notes for Experiment 9.

The gravimetric part of this experiment comes from Olmsted, J. J. *Chem. Educ.*, **1984**, 61, 1098. The spectrophotometric method comes from Dallinger, R.F. *J. Chem. Educ.*, **1995**, 72, 936.

Pre-lab Assignment for 9a:

1. Calculate the molecular weights for the following compounds (a) $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (b) $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$ (c) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ (d) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$.
(a) 403.99 (b) 184.23 (c) 491.24 (d) 437.20
2. Calculate the theoretical yield of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ assuming the masses of reagents are as written in the preparation.
moles $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O} = (3.0 \text{ g})/403.99 = 7.4 \text{ mmol}$
theoretical yield = $7.4 \times 10^{-3} \times 491.24 = 3.6 \text{ g}$
3. Dehydration in an oven does not always remove all water present. If 1.02 g dehydrated sample was obtained from 1.14 g $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$, calculate how many moles of sample went in to the oven, and how many moles of water were lost. Determine the ratio of moles of water lost per mole of sample. What is the formula of the dehydrated sample?
1.14 g $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ represents 2.32 mmol.
(1.14 - 1.02) g = 0.12 g water lost
0.12 g / 18 = 6.67 moles water
therefore ratio of water: $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] = 6.67 / 2.32 = 2.9$
Thus, we can assume that all the water has been lost.
4. Draw a flow chart of the procedure in your notebook.

Experiment 9a: In-laboratory Assignment

Group Assignment

1. Show in glorious detail, how you can arrive at the balanced equation for the titration of permanganate ion with oxalate, using the half-cell equations in the table at the back of the lab manual (0.5 marks).
 $2 \text{MnO}_4^- + 16 \text{H}^+ + 5 \text{C}_2\text{O}_4^{2-} \rightarrow 2 \text{Mn}^{2+} + 10 \text{CO}_2 + 8 \text{H}_2\text{O}$
2. Calculate how much potassium permanganate solution (assume a molarity of 0.0200 M) must be delivered to add 80% of the titre for a 0.2500 g sample of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (0.5 marks).
 $2 \text{MnO}_4^- + 16 \text{H}^+ + 5 \text{C}_2\text{O}_4^{2-} \rightarrow 2 \text{Mn}^{2+} + 10 \text{CO}_2 + 8 \text{H}_2\text{O}$
Therefore, 5 moles of oxalate are reacted with exactly 2 moles permanganate.
Since 3 moles of oxalate are furnished by each mole of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$,
then 5/3 moles of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ react with exactly 2 moles permanganate
now # mol $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ to be titrated is $0.2500 \text{ g} / 437.20 \text{ g.mol}^{-1}$
thus # mol permanganate required will be # mol $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ x 3/5 x 2 for complete reaction.

$$= 0.2500 \text{ g} / 437.20 \text{ g.mol}^{-1} \times 3/5 \times 2$$

$$= 0.686 \text{ mmol}$$

or for 80% reaction, 0.55 mmol, which translates to a volume using (# mol) / (molarity [MnO₄])

thus volume = 0.549 mmol / 0.0200 M = 27.4 mL

3. Draw the structure of [Fe(C₂O₄)₃]³⁻ in sufficient detail to show which atoms are bonded to each other and to show the geometry around the metal centre (0.5 marks).

Look for the oxalate drawn in detail and octahedral geometry.

Individual assignment for 9a:

Ratios for the individual in-lab assignment:

good: 29.85/0.2306; 31.00/0.2403; 25.63/0.2005; 30.89/0.2395; 29.08/0.2263; 26.86/0.2082.

bad: 26.61/0.2142; 30.16/0.2268; 32.48/0.2442; 29.03/0.2341.

1. Write down three volume:mass ratios given to you by the instructor. The volume represents the titre found in the oxalate vs permanganate titration, using that mass of dehydrated K₃[Fe(C₂O₄)₃] sample.
0 marks
29.85/0.2306 = 129.4; 31.00/0.2403=129.0; 25.63/0.2005=127.8;
30.89/0.2395=129.0; 29.08/0.2263=128.5; 26.86/0.2082=129.0;
26.61/0.2142=124.2; 30.16/0.2268=133.0; 32.48/0.2442=133.0;
29.03/0.2341=124.0
2. Calculate the average ratio.
0.25 marks **for correct sig figs**
3. In each case, determine the % reproducibility by applying the formula:
(value - average)/average x 100.
0.25 marks
4. A good value for percentage reproducibility is under 1% for a titration of this sort. Is a fourth titration necessary? If so, ask your instructor for a fourth data set.
5. Recalculate the average and therefore the % reproducibilities. Is there acceptable agreement now?
0.25 marks
6. Apply the Q-test to the data (you may refer to the lab manual for this) to see if any data sets can be eliminated from use in the final answer. If the answer is no, then all the data must be used in any subsequent calculations, regardless of the reproducibility of the experiment.
0.5 marks
Q test must be applied to ratios, not % reproducibilities.
7. What value of volume/mass ratio should be used in calculating the purity of the K₃[Fe(C₂O₄)₃]?
0.25 marks

This should be the average of any trials not excluded by the Q-test, regardless

of whether those data are within the reproducibility limits.

Pre-lab Assignment for 9b:

1. Calculate the molecular weights for the following compounds (a) $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (b) $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$ (c) 2,2'-dipyridyl.
(a) 179.90 (b) 392.14 (c) 156.19
2. Using the data given below, determine the mass of $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ collected.
a) mass of empty crucible = **29.0135**
b) mass of crucible + ppt = **29.1123**
c) mass of precipitate = **0.0988**
3. What is the weight of iron in the precipitate of question 2?
 $(0.0988/179.90) \times 55.85 = 0.0307 \text{ g Fe}$
4. If all of the iron of question 3 was from a 0.3025 g sample of dehydrated $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$, what would be the weight of the iron expected theoretically?
 $0.3025 \text{ g K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \text{ contains } 0.3025 \times (55.85 / 437.20) = 0.0386 \text{ g Fe}$
5. Using the answers to questions 3 and 4, calculate the percentage of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ that must have been photolysed.
Thus, $(0.0307 \text{ g}) / (0.0386 \text{ g}) = 79.5 \% \text{ photolysed}$
6. Calculate the concentration of Fe in each of the standard solutions once they have been diluted to 10 mL in the spectrometry preparation..
 $0.15 \text{ g } (\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O} \text{ in } 1 \text{ L} = (0.15 \text{ g}) / 392.14 \text{ M} = 3.825 \times 10^{-4} \text{ M}$
Thus, standard 1 (4 mL of stock) has $0.4 \times 3.825 \times 10^{-4} \text{ M} = 1.530 \times 10^{-4} \text{ M}$

2 (3 mL)	= $1.148 \times 10^{-4} \text{ M}$
3 (2 mL)	= $7.650 \times 10^{-5} \text{ M}$
4 (1 mL)	= $3.825 \times 10^{-5} \text{ M}$
7. Define a chelating ligand. Give an example from this experiment.
Oxalate is a chelating ligand because it has two points of attachment to the same metal. The dipyridyl ligand can also be cited as an example.

In-lab assignment 9b.

1. Give the oxidation states of iron in:
(a) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (0.25 marks)
III+
(b) $\text{Fe}(\text{C}_2\text{O}_4) \cdot 2\text{H}_2\text{O}$ (0.25 marks)
II+
(c) During the photodecomposition reaction, carbon dioxide is evolved. Give the half-reaction that describes how it is generated. (0.25 marks)
 $\text{C}_2\text{O}_4^{2-} \rightarrow 2 \text{CO}_2 + 2 \text{e}^-$
(d) Give the corresponding half-equation that allows this to happen. (0.25 marks)
 $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$
2. 522 nm is the wavelength chosen in this experiment? Explain how this would be determined if the value had not been supplied.

1 mark

The wavelength showing the largest absorbance for the complex is at 522 nm. This is found by scanning (plotting A vs nm) to plot a smooth curve. Ideally, the max is chosen

for this type of expt since one sees the greatest change at this point. This increases the detection accuracy. Hopefully, it does not coincide with a spectrum of starting material or other reagent!

3. Explain the point of drawing a calibration curve (0.5 marks).

This allows one to average a set of data for a series of solutions of known concentration, and then use it as a calibration device to compare those parameters with that of the unknown. The object of the exercise being to determine the concentration of the unknown.

4. Ascorbic acid is oxidised to the following structure:

(a) What causes this oxidation? (0.5 marks)

The reduction of the iron(III) to iron(II)

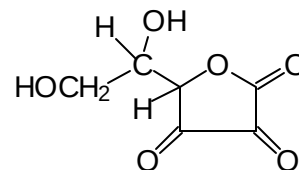
(b) Why is it necessary to have an oxidisable substance (such as ascorbic acid) present in the samples? (0.5 marks)

To make sure that all the iron is analysed, whether it is initially as iron(II) or (III).

5. Draw the structure of the complex $[\text{Fe}(\text{bipy})_3]^{2+}$.

0.5 mark

Look for a structure that is octahedral with nitrogens linking the bidentate bipy ligands (which show the appropriate detail in the carbon rings) and an overall charge of 2+.



Name:

Experiment 9a: The individual sheet for the in-laboratory assignment

1. Write down three volume:mass ratios given to you by the instructor. The volume represents the titre found in the oxalate vs permanganate titration, using that mass of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ sample.

	I	II	III
volume			
mass			
volume/mass =			

2. Calculate the average ratio (0.25 marks).

3. In each case, determine the % reproducibility by applying the formula (0.25 marks):
(value - average)/average x 100.

	I	II	III
% reproducibility			

4. A good value for percentage reproducibility is under 1% for a titration of this sort. Is a fourth titration necessary? If so, ask your instructor for a fourth data set.

	IV
volume	
mass	
volume/mass =	

New average =

5. Recalculate the average and therefore the % reproducibilities. Is there acceptable agreement now (0.25 marks)?

	I	II	III	IV
% reproducibility				

6. Apply the Q-test to the data (refer to the lab manual for this) to see if any data sets can be eliminated from use in the final answer. If the answer is no, then all the data must be used in any subsequent calculations, regardless of the reproducibility of the experiment (0.5 marks).
7. What value of volume/mass ratio should be used in calculating the purity of the $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (0.25 marks)?

Names of the group members:

Experiment 9a: The group sheet for the in-laboratory assignment

1. Show in glorious detail, how you can arrive at the balanced equation for the titration of permanganate ion with oxalate, using the half cell equations in the table at the back of the lab manual (0.5 marks).
2. Calculate how much potassium permanganate solution (assume a molarity of 0.020 M) must be delivered to add 80% of the titre for a 0.2500 g sample of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (0.5 marks).
3. Draw the structure of $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ in sufficient detail to show which atoms are bonded to each other and to show the geometry around the metal centre (0.5 marks).

Names of the group members:

Experiment 9b: In-laboratory Group Assignment

1. Give the oxidation states of iron in:

(a) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (0.25 marks)

(b) $\text{Fe}(\text{C}_2\text{O}_4) \cdot 2\text{H}_2\text{O}$ (0.25 marks)

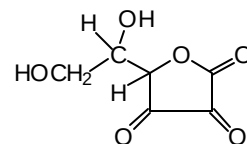
(c) During the photodecomposition reaction, carbon dioxide is evolved. Give the half-reaction that describes how it is generated. (0.25 marks)

(d) Give the corresponding half-equation that allows this to happen. (0.25 marks)

2. 522 nm is the wavelength chosen in this experiment? Explain how this would be determined if the value had not been supplied (1 mark).

3. Explain the point of drawing a calibration curve (0.5 marks).

4. Ascorbic acid is oxidised to the following structure:



- (a) What causes this oxidation? (0.5 marks)
- (b) Why is it necessary to have an oxidisable substance (such as ascorbic acid) present in the samples? (0.5 marks)
5. Draw the structure of the complex $[\text{Fe}(\text{bipy})_3]^{2+}$ (0.5 marks).